

Math 534H - Spring 2026 - Lecture 19

①

From last time we got:

$$\frac{\overbrace{T''(t)}^{\text{dep. only on } t}}{\underbrace{c^2 T(t)}_{\text{dep. only on } x}} = \frac{\overbrace{X''(x)}^{\text{dep. only on } x}}{\underbrace{X(x)}_{\text{dep. only on } x}} = -\lambda = \text{constant.}$$

$\Rightarrow$ can only be constant.

Let's call this Constant: $-\lambda$ and

we assume (for now; later we will prove it)

that $\lambda > 0$ $\leftarrow$ CLAIM TO BE PROVEN
LATER.

So let's assume for now and continue.

This give us two second order ODE's : ②

$$\begin{cases} T''(t) = -\lambda c^2 T(t) & \forall t > 0 \\ X''(x) = -\lambda X(x) & , \forall x \in [0, L] \end{cases}$$

Need to solve these two ODE ; for

convenience let us call $\lambda = \beta^2 \Leftrightarrow \beta := \sqrt{\lambda}$
(using $\lambda > 0$)

We have

$$\begin{cases} T''(t) + \beta^2 c^2 T(t) = 0 \\ X''(x) + \beta^2 X(x) = 0 \end{cases}$$

Aside from ODE's : to look at ODE's we

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look at the roots of $r^2 + \beta^2 = 0$ (for X)

has complex roots

$$\text{If } r^2 = -\beta^2 \Rightarrow \begin{cases} r_1 = i\beta \rightarrow e^{i\beta x} \\ r_2 = -i\beta \rightarrow e^{-i\beta x} \end{cases}$$

Recall

$$e^{i\beta x} = \cos(\beta x) + i \sin(\beta x)$$

$$\begin{aligned} e^{-i\beta x} &= \cos(-\beta x) + i \sin(-\beta x) \\ &= \cos(\beta x) - i \sin(\beta x). \end{aligned}$$

This tells us that the solutions of the ODE's ④
are :

$$X(x) = C \cos(\beta x) + D \sin(\beta x)$$

for any C, D constants.

$$T(t) = A \cos(\beta \cdot c \cdot t) + B \sin(\beta \cdot c \cdot t)$$

for any A, B constants.

But now recall we have zero Dirichlet

Conditions : $u(0, t) = 0 = u(L, t) \quad \forall t \geq 0$

We have $u(x,t) = X(x) \cdot T(t)$. So ⑤

$$u(0,t) = 0 = X(0)T(t) \quad \forall t > 0$$

$$u(L,t) = 0 = X(L)T(t) \quad \forall t > 0$$

$$\Rightarrow \boxed{X(0) = 0 = X(L)}.$$

Let's plug this in in our formula for $X(x)$

$$0 = X(0) = C \Rightarrow \boxed{C = 0} \text{ then}$$

$$\underline{X(x) = D \sin(\beta x)}; \text{ but we also}$$

$(u \equiv 0 \quad \leftarrow D \neq 0 \text{ otherwise } X \equiv 0)$

(6)

know that

$$0 = X(L) = D \cdot \sin(\beta \cdot L), \quad D \neq 0$$

$$\Rightarrow \sin(\beta L) = 0 \quad \Rightarrow$$

$\uparrow$
 given

$$\beta L = n\pi \quad \text{for any integer } n$$

$$\Rightarrow \boxed{\beta = \frac{n\pi}{L}} \quad \text{for any integer } n.$$

(wlog, $n \geq 1$).

Hence

$$\boxed{X_n(x) = D \sin\left(\frac{n\pi}{L} x\right)}$$

↳ also can change with n (later)

and

(7)

$$T_n(t) = A_n \cos\left(\frac{n\pi}{L} \cdot ct\right) + B_n \sin\left(\frac{n\pi}{L} ct\right)$$

Hence for every n integer; $n \geq 1$

$$u_n(x, t) = X_n(x) \cdot T_n(t)$$

$$= D \cdot \sin\left(\frac{n\pi}{L} x\right) \cdot \left[A_n \cos\left(\frac{n\pi}{L} ct\right) + B_n \sin\left(\frac{n\pi}{L} ct\right) \right]$$

Let's relabel the constants as $\tilde{A}_n = D A_n$

$$\tilde{B}_n = D \cdot B_n.$$

A_0 :

for any $n \geq 1$

(8)

$$u_n(x, t) = \left[\tilde{A}_n \cos\left(\frac{n\pi}{L} ct\right) + \tilde{B}_n \sin\left(\frac{n\pi}{L} ct\right) \right] \sin\left(\frac{n\pi}{L} x\right)$$

where $\tilde{A}_n, \tilde{B}_n$ are arbitrary constants, $n \geq 1$

→ BUILDING BLOCKS for any solution to

$$(*) \begin{cases} u_{tt} - c^2 u_{xx} = 0 & x \in [0, L], t > 0 \\ u(0, t) = 0 = u(L, t), & t > 0 \end{cases}$$

In particular $u_N(x, t) = \sum_{n=1}^N u_n(x, t)$ is

also a solution to $(*)$; for any arbitrary $N \geq 1$ ⑨
(by finite superposition of solutions).

A question for later : Does the limit as $N \rightarrow \infty$
of $z_N(x, t)$ exist? If so, does it give
rise to a solution?

$$\lim_{N \rightarrow \infty} z_N(x, t) = \sum_{n=1}^{\infty} z_n(x, t)$$

Now let's return to the INITIAL CONDITIONS
that we were given :

$$u(x,0) = \phi(x) \quad ; \quad u_t(x,0) = \psi(x)$$

(given) (given).

QUESTION When is it true that :

$$u_N(x,0) = \sum_{n=1}^N \tilde{A}_n \sin\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \underbrace{\phi(x)}_{\text{given}}$$

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For example if $\phi(x) = \sin x$; $L = 2\pi$ we have that for

$$\tilde{A}_n = \begin{cases} 1 & n=2 \\ 0 & n \neq 0 \end{cases}$$

holds.

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